

UX/UI Course Capstone · Pre-Design Phase

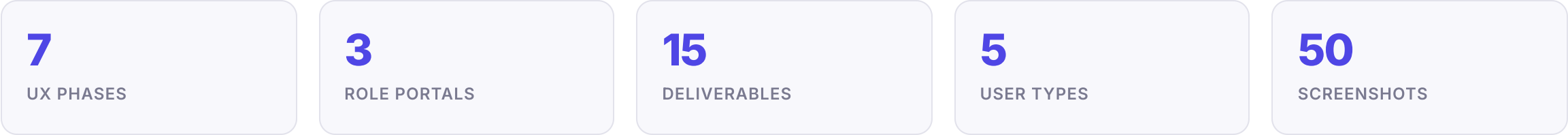
# Travelex

## T&E Platform

### PROCESS DOCUMENTATION

A structured, AI-assisted UX methodology from problem definition to implementation-ready handoff.

# Project at a Glance



| ATTRIBUTE           | DETAIL                                                        |
|---------------------|---------------------------------------------------------------|
| Project Type        | UX/UI Capstone — Pre-Design Phase                             |
| User Types          | Employee · Manager · Finance · Admin · Executive/CFO          |
| Portals Implemented | Employee · Manager · Finance (Admin + Executive consolidated) |
| Key Frameworks      | Kano Model · MoSCoW · JTBD · RBAC · ERD · Journey Mapping     |
| Handoff             | Full-stack Travelex app + NotebookLM knowledge base           |

# The Business Problem

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T&E management runs on email threads, paper receipts, and spreadsheets at most companies. The tension is structural — four groups, four different definitions of "done."

## WHO SUFFERS

- **Employees** — lose receipts, no visibility on where their reimbursement is
- **Managers** — approval bottlenecks, no policy context, can't delegate
- **Finance** — manual GL coding, violations caught after the fact
- **Executives** — spend data arrives weeks late, no proactive alerts

## WHAT TRAVELEX SOLVES

- Pre-trip travel approval with budget tracking
- Expense capture with AI-assisted GL coding
- Multi-tier approval workflow with policy flags at submission time
- Real-time spend analytics for finance and management

# The Six-Step Methodology

Each phase was treated as a gate — outputs had to be sufficient before proceeding. Iteration happened *within* each step, not across them.

| # | PHASE                    | KEY OUTPUT                                                   |
|---|--------------------------|--------------------------------------------------------------|
| 1 | Problem Definition       | Bounded problem statement, scope decisions, success criteria |
| 2 | Research                 | Market research report, user sentiment synthesis             |
| 3 | Personas & User Stories  | 5 personas with JTBD framing and user stories                |
| 4 | Journey Mapping          | 3 end-to-end flows, cross-role handoff diagram               |
| 5 | Feature Prioritization   | Kano + MoSCoW matrix, MVP scope locked                       |
| 6 | Information Architecture | Sitemap, user action tree, wireframe prompts                 |

*Every deliverable was drafted by AI, critiqued by the designer, and iterated — almost nothing was accepted on first pass.*

# 01

## Problem Definition

Framing the T&E problem, setting scope boundaries, and defining measurable success criteria before any requirements were written.

# Problem Definition

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## IN SCOPE

- Travel pre-approval requests
- Expense capture and submission
- Manager approval workflow
- Finance GL coding and policy enforcement
- Spend reporting and analytics

## EXPLICITLY OUT OF SCOPE (MVP)

- In-app travel booking engine
- Corporate card auto-reconciliation
- ERP / accounting system integration
- Procurement and vendor payments

## Success criteria defined:

- Reimbursement cycle time reduction
- Policy compliance rate improvement
- Finance month-end close time reduction

*Excluding a booking engine was the single most important scope decision — it would have doubled the technical surface area and shifted the product toward a travel marketplace model.*

# 02

## Research

Market analysis of the T&E SaaS landscape and synthesis of user sentiment from public sources.

# Research Findings

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## WHAT INCUMBENTS DO WELL

- Approval workflow routing
- OCR receipt capture
- Corporate card integration
- Policy rule engines

## WHERE THEY FALL SHORT

- Employee submission experience — an afterthought
- Over-complex admin configuration
- Weak real-time analytics for executives
- Policy feedback at rejection, not submission

### Design differentiator identified:

The employee-side experience is consistently rated lowest in reviews. Most tools are optimised for Finance — the person submitting the expense is a secondary consideration. Positioning Travelex around a **fast, low-friction submission flow with policy guidance at point of entry** became the core design principle carried forward.



# 03

## Personas & User Stories

Five user types grounded in real frustration patterns, each with JTBD framing and representative user stories.

# Persona — Employee

## Alice · Engineering

Submits 8–12 expenses per month · High travel frequency

| Primary goal    | Submit quickly, get reimbursed without friction                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Key frustration | Policy rules are unclear at submission; rejection arrives days later with no explanation |
| JTBD            | <i>Hire the platform to make submitting expenses so fast it becomes a non-event</i>      |

### User stories:

- As an employee, I want to photograph a receipt and have the amount pre-filled, so I can submit in under 60 seconds.
- As an employee, I want to see where my report is in the approval chain, so I can follow up with the right person.
- As an employee, I want a clear rejection reason, so I can resubmit without going back and forth by email.

# Persona — Manager

## Mike · Engineering Manager

Reviews 15–25 expense reports per month · Frequently travelling

|                 |                                                                                |
|-----------------|--------------------------------------------------------------------------------|
| Primary goal    | Approve accurately and quickly without becoming a bottleneck                   |
| Key frustration | Approval requests arrive through email with no policy context                  |
| JTBD            | <i>Hire the platform to handle approval overhead so I can focus on my team</i> |

### User stories:

- As a manager, I want a policy violation summary before I approve, so I can decide without reading every line item.
- As a manager, I want to delegate approval authority when I am out of office, so reimbursements are not blocked.
- As a manager, I want a daily digest of pending approvals — not one email per request.

# Persona — Finance

## Sarah · Finance Manager

Owns GL coding, policy enforcement, and month-end close

|                 |                                                                                                  |
|-----------------|--------------------------------------------------------------------------------------------------|
| Primary goal    | Close the books accurately and on time; produce audit-ready records                              |
| Key frustration | Approved reports arrive requiring manual GL coding; violations caught too late                   |
| JTBD            | <i>Hire the platform to eliminate manual reconciliation and give me confidence in compliance</i> |

### User stories:

- As a finance team member, I want AI-suggested GL codes pre-populated on approved reports, so I can confirm rather than assign from scratch.
- As a finance team member, I want a dedicated coding queue for batch processing of transactions.
- As a finance team member, I want spend-by-category reports filterable by department and date range.

## Personas — Admin & Executive

### ADMINISTRATOR

IT / Operations · Configures the platform

- **Goal:** Configure policies and users without engineering support
- **Frustration:** Policy changes require IT tickets in legacy tools
- **JTBD:** *Hire the platform to run itself with minimal maintenance*

**Implementation note:** Admin functions merged into the Finance role in Travelex.

### EXECUTIVE / CFO

C-Suite · Needs real-time spend visibility

- **Goal:** Real-time spend visibility; identify budget risks early
- **Frustration:** Spend data arrives at month-end, no proactive alerts
- **JTBD:** *Same visibility into T&E spend as into revenue*

**Implementation note:** Executive analytics live under Manager → Team in Travelex.

# 04

## Journey Mapping

End-to-end flows mapped across roles — with the handoffs between them made explicit.

# Journey Mapping — Key Insight

Three flows mapped: **Employee submission → Manager review → Finance processing.**

The most valuable output was making cross-role handoffs explicit:

| HANDOFF POINT | FROM → TO          | WHAT PASSES                        | FAILURE MODE                                     |
|---------------|--------------------|------------------------------------|--------------------------------------------------|
| Submission    | Employee → Manager | Report + line items + policy flags | Manager has no context to decide quickly         |
| Approval      | Manager → Finance  | Approved report + review note      | Finance still needs to manually assign GL codes  |
| Coding        | Finance → Payment  | GL - coded report                  | Report sits in queue; employee has no visibility |
| Rejection     | Manager → Employee | Review note                        | No explanation → incorrect resubmission          |
| Escalation    | Manager → Finance  | High - value report                | No threshold - based routing                     |

*The moment a manager approves a report and it enters the finance queue is **completely invisible to the employee** — this is why "where is my reimbursement?" is the most common T&E complaint. This surfaced a core requirement: a timestamped audit trail accessible to employees at all times.*

# 05

## Feature Prioritization

Dual-framework approach: Kano Model for user value classification, MoSCoW for release scoping.



# Kano + MoSCoW Matrix

| FEATURE                                  | KANO        | MOSCOW      | RELEASE   |
|------------------------------------------|-------------|-------------|-----------|
| Receipt capture (OCR photo-to-form)      | Basic       | Must Have   | MVP       |
| Multi-tier approval workflow             | Basic       | Must Have   | MVP       |
| Policy rule engine (per-category limits) | Basic       | Must Have   | MVP       |
| Reimbursement status tracking            | Performance | Must Have   | MVP       |
| Mobile-first submission flow             | Performance | Must Have   | MVP       |
| Real-time spend dashboard                | Performance | Should Have | MVP       |
| AI-suggested GL code assignment          | Excitement  | Should Have | MVP       |
| Delegation of approval authority         | Performance | Should Have | MVP       |
| Bulk CSV export                          | Basic       | Must Have   | MVP       |
| Anomaly / duplicate receipt detection    | Excitement  | Could Have  | Release 2 |
| ERP integration                          | Basic       | Won't Have  | Release 2 |
| In-app travel booking engine             | Indifferent | Won't Have  | Release 3 |

# 06

## Information Architecture

Sitemap, RBAC-shaped user action tree, and wireframe prompts — the handoff layer between discovery and visual design.

# Portal Navigation — Implemented

RBAC principles were used as an IA-shaping tool: mapping create/read/update/delete permissions per role per entity determined what appeared in each portal's navigation.

| PORTAL   | NAVIGATION                                                                                      |
|----------|-------------------------------------------------------------------------------------------------|
| Employee | Dashboard · Expenses · Travel · Settings                                                        |
| Manager  | Dashboard · Approvals · Expenses · Travel · Team (+ Executive analytics) · Analytics · Settings |
| Finance  | Dashboard · Policies · Expenses · Travel · Coding Queue · Analytics · Export · Settings         |

**Key IA decision:** The **GL Coding Queue** is a dedicated sequential interface — not embedded in the expense list. Finance team members process transactions in batches; context-switching between review and coding is a known source of errors. This decision came directly from the Finance journey map.

# Data Model — ERD Entities

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The ERD was produced during the pre-design phase — not delegated to engineering — because data model decisions have direct UI consequences.

## CORE ENTITIES (IMPLEMENTED)

- **User** — role: employee / manager / finance
- **ExpenseReport** — status: draft → submitted → under\_review → approved → rejected → paid
- **ExpenseItem** — category, amount, receipt URL, policy violation flag
- **TravelRequest** — destination, dates, budget, status, reviewer

## SUPPORTING ENTITIES

- **PolicyRule** — per-category spending limits and receipt thresholds
- **Notification** — typed, user-targeted alerts with read state

Note: Pre-design ERD anticipated a separate *Approval* entity. Implementation embedded approval state as fields on ExpenseReport and TravelRequest — simpler and sufficient for the access patterns needed.

07

## Visual Design Phase

Modern design system applied to the implemented platform —  
frosted glass cards, indigo brand, animated data visualization.

# Design System — Tokens & Aesthetic

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## DESIGN TOKENS

- **Brand:** Indigo `#4F46E5` ( `--brand-600: 79 70 229` )
- **Light surface-0:** faint blue-tinted canvas `#F8F8FC`
- **Light surface-1:** white cards `#FFFFFF`
- **Dark surface-0:** true charcoal `#121212`
- **Dark surface-1:** near-black cards — high contrast

## APPLE / CRAFT AESTHETIC

- **.card:** `bg-white/80 backdrop-blur-xl border border-black/7 rounded-[14px] shadow`
- **.glass:** sidebar + header — `bg-white/75 backdrop-blur-xl`
- **Font:** Inter — SF Pro-inspired weight + tracking
- **Labels:** `10px semibold uppercase tracking-[0.1em]`
- **Sidebar subtitle:** "EXPENSE INTELLIGENCE"

# Dashboard — Bento KPI Grid

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## LAYOUT & CHARTS

- Bento asymmetric layout — Total Spend spans 2 cols
- KPI cards with accent strip + mini sparkline + delta % badge
- Period toggle 7d / 30d / 90d filters trend chart
- Gradient area chart with animated entry
- Illustrated empty states with CTA
- Hover arrow animations on expense/travel rows

## NEW COMPONENTS BUILT

- **Sparkline** — mini AreaChart, no axes, gradient fill
- **Delta** — green/red trend badge with icon
- **KpiCard** — accent strip + sparkline + delta
- **ChartTooltip** — custom dark/light tooltip
- **PeriodToggle** — 7d/30d/90d pill buttons
- **EmptyState** — icon + message + CTA

# Sidebar — Collapsible + Live Counts

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## SIDEBAR UPGRADES

- Collapsible to 60px icon-only mode (localStorage persisted)
- Live pending count badges on nav items — dot on icon when collapsed, pill when expanded
- "EXPENSE INTELLIGENCE" branding (small-caps, letter-spaced)
- Smooth CSS transition: 220px ↔ 60px
- Collapse toggle button at bottom

## FORM & TABLE UPGRADES

- 2-column form layout with narrow Tips card on the right
- Tips card: indigo top border + "Tips" heading in brand color
- Uppercase field labels matching Figma spec
- Department dropdown + Description textarea added
- Sortable column headers on all tables (click to sort asc/desc)
- **SortIcon** — ChevronUp/Down/ChevronsUpDown indicator



# Analytics — Upgraded Data Visualization

| CHART          | BEFORE              | AFTER                                              |
|----------------|---------------------|----------------------------------------------------|
| Monthly Trend  | LineChart (no fill) | AreaChart + gradient fill + animated entry         |
| Category Split | PieChart (flat)     | Donut chart + legend list                          |
| Department     | Horizontal bar      | Gradient horizontal bar (indigo to lighter indigo) |
| All charts     | Basic tooltip       | Custom dark/light styled tooltip                   |
| Trend chart    | Fixed time window   | Period toggle (7d / 30d / 90d)                     |

**Additional details:** rounded bar tops on all bar charts ( `radius= {[4,4,0,0]}` ); donut `innerRadius` added to PieChart; linearGradient left-to-right on department bars; KPI cards upgraded with colored icons + delta badges.

# Modern UX — Research-Driven Upgrades

Before implementing, competitive research was conducted across Ramp, Brex, Navan, Expensify, Airbase, and Spendesk — identifying 10 modern patterns to implement.

| INSPIRATION    | PATTERN IMPLEMENTED                  |
|----------------|--------------------------------------|
| Ramp / Brex    | Bento asymmetric KPI grid            |
| Navan          | Live pending badge counts on sidebar |
| Brex           | Sparklines + delta % on KPI cards    |
| Ramp / Airbase | Period toggle on analytics           |
| Spendesk       | Gradient fills under area charts     |
| Expensify      | Illustrated empty states with CTA    |
| Ramp           | Sortable column headers              |
| Figma designs  | 2 - column form layout + Tips card   |

# Deliverables Produced

| DELIVERABLE                                     | PURPOSE                                                                      |
|-------------------------------------------------|------------------------------------------------------------------------------|
| Problem statement                               | Scope boundary, success criteria — north - star reference                    |
| Market research report                          | Competitive context; ingested into NotebookLM                                |
| Five personas + user stories                    | Foundation for all downstream design decisions                               |
| Journey maps (3 flows)                          | Cross - role handoff visibility; surfaced status - tracking requirement      |
| Kano + MoSCoW matrix                            | MVP scope — defensible to design and engineering audiences                   |
| Sitemap + user action tree                      | Navigation structure per role; source of truth for RBAC                      |
| ERD                                             | Backend data model; produced pre - design not post                           |
| Wireframe prompts (3 portals)                   | Primary handoff artifact for visual design phase                             |
| <b>Travelx — full - stack implementation</b>    | React + FastAPI + SQLite; 3 role - based portals                             |
| Consolidated export + NotebookLM corpus         | Queryable design knowledge base                                              |
| <b>Visual design system + component library</b> | Frosted glass cards, 7 reusable components (Sparkline, Delta, KpiCard, etc.) |
| <b>50 UI screenshots (1440×900)</b>             | Full coverage: 3 roles × all pages, light + dark mode                        |

# AI-Assisted Workflow — The Iteration Loop

## WHERE AI EXCELLED

- **Structural scaffolding** — first-draft frameworks and templates at full depth
- **User story generation** — grammatically and structurally correct at scale
- **Diagram syntax** — valid Mermaid / ERD notation reliably
- **Edge case surfacing** — volunteered scenarios not explicitly considered
- **Consistency enforcement** — flagged decisions that contradicted prior outputs

## WHERE HUMAN JUDGMENT WAS ESSENTIAL

- **Deciding which users matter** — Admin vs. Finance sub-role required domain judgment
- **Grounding personas** — AI defaults are optimistic; real frustrations needed pushing
- **MVP scoping** — tradeoffs no framework can make automatically
- **Navigability validation** — logical grouping ≠ navigability; required UX judgment to flatten

*The standard loop: seed context → AI generates draft → designer critiques → iterate → lock deliverable. Almost nothing was accepted on first pass.*

# Outcomes & Lessons Learned

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## WHAT WORKED WELL

- **Sequential phase discipline** — each phase as a gate produced more defensible later-phase outputs
- **Cross-role journey mapping** — the single most productive methodological decision
- **RBAC as IA principle** — applied before design began, not retrofitted
- **Iterative critique** — resisting first-pass acceptance separated grounded from generic

## WHAT TO CHANGE NEXT TIME

- **Structured competitive benchmark** — feature comparison table vs. incumbents
- **Earlier ERD** — data model as input to feature prioritization, not output of IA
- **User testing proxies** — at least one validation round with real users in target roles

**Skills reinforced:** Scope definition before requirements · Complementary framework pairing (Kano + MoSCoW) · Cross-role handoff thinking · Pre-design data modelling

Travelex — Expense Intelligence Platform

# Thank you

UX/UI Capstone · Pre-Design Phase Documentation

3 role-based portals · 7 phases · 15 deliverables · 50 screenshots

Built with Claude · React · FastAPI · Travelex